

Definitions:

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Examples:

Theorem. Let G be a group and let $a \in G$.

Corollaries:

- 1.

- 2.

- 3.

Theorem. Let a be an element of order n in a group G .

Examples:

Corollaries:

1.

2.

3.

4.

Question 1. What are the generators of $\mathbb{Z}_{10}$?

Question 2. Let $(G, *)$ be a cyclic group of order 12 with generator a . Determine the order of a^2 , a^7 , and a^8 . Are there any other generators of G ?

Theorem (Fundamental Theorem of Cyclic Groups).

Question 3. Let a and b be elements of a group with $|a| = 10$ and $|b| = 21$. Why does the intersection of $\langle a \rangle$ and $\langle b \rangle$ only contain the identity of the group?

Question 4. How many subgroups are there of $\mathbb{Z}_{36}$? What are they?

Question 5. If G is an Abelian group and contains cyclic groups of orders 4 and 6, what other sizes of cyclic subgroups must G contain?

Euler phi function, $\phi(d)$:

Theorem. If d is a positive divisor of n , then

Proof:

Corollary.